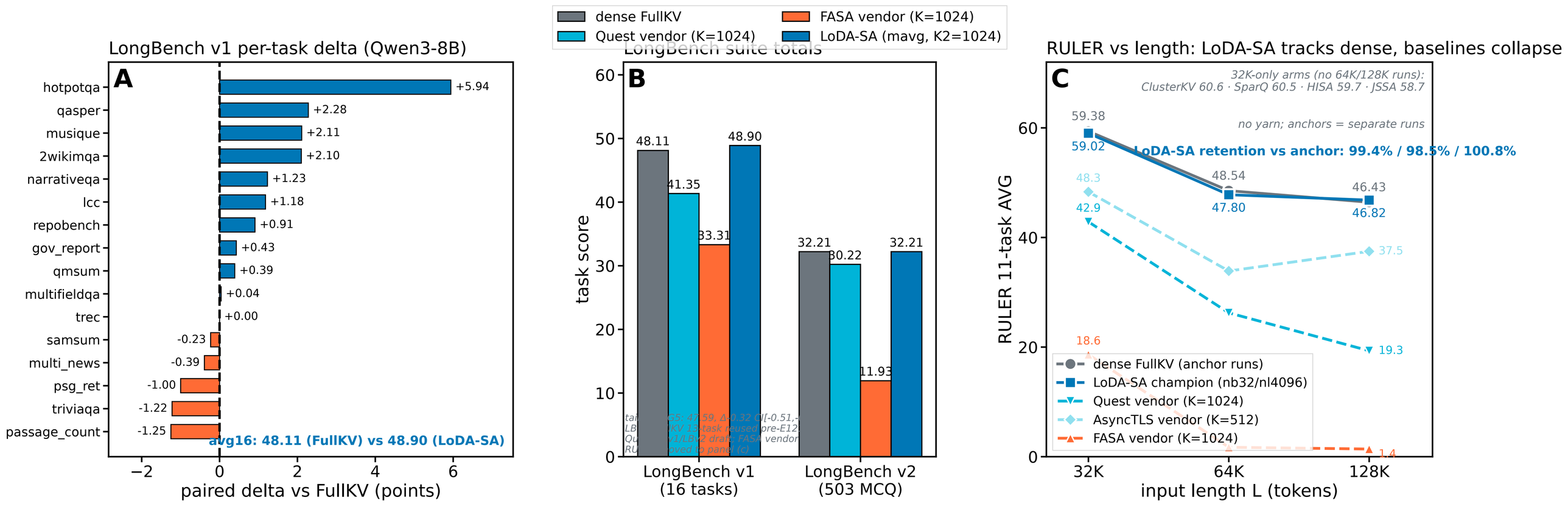
dense FullKV
Quest vendor (K=1024)
FASA vendor (K=1024)
LoDA-SA (mavg, K2=1024)
A
LongBench v1 per-task delta (Qwen3-8B)
hotpotqa +5.94
qasper +2.28
musique +2.11
2wikimqa +2.10
narrativeqa +1.23
lcc +1.18
repobench +0.91
gov_report +0.43
qmsum +0.39
multifieldqa +0.04
trec +0.00
samsum -0.23
multi_news -0.39
psg_ret -1.00
triviaqa -1.22
passage_count -1.25
avg16: 48.11 (FullKV) vs 48.90 (LoDA-SA)
paired delta vs FullKV (points)
B
LongBench suite totals
task score
48.11
41.35
33.31
48.90
32.21
30.22
11.93
32.21
LongBench v1
(16 tasks)
LongBench v2
(503 MCQ)
C
RULER vs length: LoDA-SA tracks dense, baselines collapse
32K-only arms (no 64K/128K runs):
ClusterKV 60.6 · SparQ 60.5 · HISA 59.7 · JSSA 58.7
no yarn; anchors = separate runs
LoDA-SA retention vs anchor: 99.4% / 98.5% / 100.8%
59.38
59.02
48.54
47.80
46.43
46.82
48.3
42.9
37.5
19.3
18.6
1.4
dense FullKV (anchor runs)
LoDA-SA champion (nb32/nl4096)
Quest vendor (K=1024)
AsyncTLS vendor (K=512)
FASA vendor (K=1024)
RULER 11-task AVG
32K
64K
128K
input length L (tokens)